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Early Real Estate Indicators during the COVID-19 Crisis

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In this article we construct a number of early housing-market indicators from daily-scraped list-price data and investigate how they inform on housing-market trends prior to the release of official transaction data. We minimize the impact of non-market conform list prices by our data selection- and cleaning routine and thereby improve the overlap between list and transaction data. In addition to some well-known real-estate indicators (price and rent indices, price-rent ratios, market volume, *time-on-market*, and market turnover) we develop a new market sentiment indicator which we construct from the direction and frequency of online price changes for individual listings. We then use this group of indicators to investigate how housing markets in London and Vienna react to the COVID-19 pandemic during 2020. For London these indicators show high volatility during much of 2020 and we find that the sales and rental markets drift apart: The sales market shows an overall positive price trend, while the rental market becomes significantly weaker after the onset of the COVID-19 crisis. For Vienna we find that all indicators signal positive market developments for the first eight months after the start of the COVID-19 pandemic.

Key words: List price index; COVID-19; real-estate indicators; online sentiment indicator; leading indicators.

1. Introduction

For government agencies and central banks, it is important to be informed about the housing market, as housing market crises can have serious consequences for the rest of the economy and threaten the banking system via mortgage defaults (see e.g., discussion in Mishkin 2011). For this reason, National Statistical Agencies, as well as National Banks, compute a number of indicators to track the development of the housing market. House price indices are the most important of them.

In official statistics, the preferred option to track the development of housing prices is via price indices that are constructed with transaction data (Eurostat 2013), but the time-lags involved in processing transaction data pose obstacles to the timely production of real estate indices and indicators. Several months can pass from a property being listed until an offer is agreed upon and several more until contracts are exchanged, completed, and entered into land registries. The typical time-lag between the availability of offer and transaction prices varies between housing markets; for example, Shimizu et al. (2016) find an average time-lag of seven months between list and transaction prices for apartment buildings in Tokyo, Brounen and Kok (2011) find an average lag of four, and a half months

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for the Netherlands, and a Bank of England estimate (presented in Eurostat 2013, 103) indicates a five, and a half months lag in the United Kingdom (UK). The implication of this time-lag is that governments and central banks often need to announce policies before official housing market indicators are available. This matters particularly during periods of crisis, such as the COVID-19 pandemic.

Real-estate data from online platforms are a potential alternative to transaction data for measuring developments in the housing market. Such list-price data have multiple benefits: They are immediately available, continuously updated, and often much cheaper to access than transaction prices. In addition to information on prices, they provide a range of other information that typically far exceeds the characteristics available in transaction data (Shimizu et al. 2016; Kolbe et al. 2021). Such listed housing data also allow for insights into market dynamics that are not available from transaction-based data sets. For example, frequent scraping enables the tracking of individual properties over time.

Some researchers have already demonstrated the usefulness of online data for spotting early signs of housing market changes. The work by Choi and Varian (2012), Chauvet et al (2014), Wu and Brynjolfsson (2015), or Lyons (2019) fall into this category. Similarly, Van Dijk and Francke (2018) illustrate with Dutch data that market tightness indicators built from online search data (online *list volume* and number of clicks per postcode) are able to predict changes in transaction prices and market liquidity. Others find list-price data less reliable for subsequent housing market development (see e.g., Knight et al. 1994; Kolbe et al. 2021).

The relationship between list and transaction prices is not straightforward. Sellers are more reluctant to lower prices during a market down-turn than increase them during a market rise (Genesove and Mayer 2001). As a result, the gap between list-prices and transaction prices varies anti-cyclically, and during housing booms, transaction prices can even exceed asking prices (for discussion and empirical examples, see e.g. Haurin et al. 2013). Additionally, the spread between list-prices and transaction prices also tends to be higher at the upper end of the market, which further complicates the relationship. This tendency can be explained by sellers' uncertainty over buyers' willingness to pay, which tends to be higher for atypical and highend properties (Haurin et al. 2010). The higher spread for high-end properties can also be explained by market liquidity. This inverse relationship between market liquidity and the price spread is wellknown in financial markets (see e.g., Bagehot 1971). And finally, each market is different, which implies that evidence from one market cannot easily be transferred to others. The regulatory environment and real-estate agent incentives also play a role in how list prices are pitched to the marketplace.

In this article we present a collection of leading indicators and apply them to two distinct housing markets: London and Vienna. Our sentiment indicator, constructed from the relative number of positive and negative price changes occurring in the market – is new to the literature. The other indicators are well known, but list-price data allow us to track them in a continuous fashion throughout the housing cycle.

We build our list-price data sets for both cities by scraping data from leading online housing platforms. Our list-price data run from the end of 2018 to the end of 2020. For 2019 we also have access to transaction data for both cities and can illustrate the relationship between the two data distributions (see Section 3). We show that our data preparation routine, which consists of excluding extreme values and limiting each listed property to

enter the data set only once (the last entry before it leaves the market), moves the price distribution of the list-price data set towards that of the transaction price data set. We believe that this closeness in price distributions translates into more accurate leading indicators.

However, our main aim in this article is not to investigate the relationship between transaction prices and list prices per se, but to illustrate how to derive list price indicators that can be used to "feel the temperature" of the market before official transaction prices are available. We do not suggest that the levels of these indicators should be interpreted as substitutes to their transaction-based counterparts. Rather, we see these indicators as performing the role of lights on a dashboard that informs on the general market direction before transaction data become available: The more indicators align in their message, the stronger the message on market tendency.

The pictures these indicators paint for the London and Vienna housing markets in 2020 after the start of the COVID-19 crisis in February/March are as follows: Neither London nor Vienna experienced a negative price development between February and December 2020. This is surprising, since previous research has shown that – at least in the short run – pandemics tend to have a negative impact on house prices (Wong 2008; Francke and Korevaar 2021). There are however significant differences in the development of the London and Vienna real estate markets during 2020.

In Vienna, both the sales and rental market exhibited all signs of "hot" property markets throughout 2020 in the form of increasing (list) prices and increasing price-rent ratios. Liquidity and market sentiment levels (for the sales as well as the rental market) are stable throughout 2020 – but at higher levels than during 2019.

The London property market indicators behaved more nervously after the start of the COVID-19 pandemic: The *list-price index* development for 2020 was stable in the sales market but negative in the rental market, and list-price volatility increased in the sales as well as the rental market after the start of the COVID-19 crisis. Volatility is also visible in the development of the liquidity indicators: *list volume*, *list time-on-market*, and *list turnover rate*. However, after the initial drop in liquidity during the first lock-down period in April 2020, liquidity increased for the second half of 2020. Finally, the *sentiment indicator* suggests a stable tendency in the sales market but a slightly negative (and quite volatile) trend in the London rental market for much of 2020.

2. List-data Indicators

We compute and discuss the following six list-data indicators:

1. List-price index as well as a list-rent index
2. List price-rent ratio
3. List volume
4. List time-on-market
5. List turnover rate
6. Market sentiment indicator based on list price changes

Transaction prices are the result of negotiations between buyers and sellers, and the *price indices* (or *rent indices*) constructed from them measure the development of these market prices over time. List-prices (list-rents), on the other hand, illustrate the market perception of sellers

(landlords) only, and the resulting *list-price indices* (*list-rent indices*) measure how this perception changes over time. The direct impact of the demand side of the market is missing in list-data and as a result, the relationship between list and transaction prices is not straightforward.

As sellers are typically slower to adjust their price ideas in a downward market than in an upward one (Genesove and Mayer 2001), we do not expect the *list-price index* to show negative price movements as fast as positive price movements. This reluctance by sellers to reduce prices during a market down-turn also implies that list-prices are not always good indicators of market turning points (Knight et al. 1994). This asymmetry in list-price reaction to underlying market tendency probably explains why the literature provides a mixed verdict on whether list prices are good leading indicators for transaction prices: While some researchers find them to be good leading indicators (see e.g., Ahmed et al. 2016; Anenberg and Laufer 2017; Lyons 2019) others find the opposite (Kolbe et al. 2021).

The *price-rent ratio* is the ratio of what a property sells for relative to its rental price. When constructed with list-price data, the *list-price ratio* answers the question of how the list price of a property relates to the annual list rent. We again only use the last price entries (for sales as well as rental properties) to construct this ratio.

It is interesting to monitor the evolution of the price-rent ratio in housing markets, as it is an indicator of whether housing markets are valued fairly or in a bubble. The relative velocity and magnitude of price changes in the housing market also leads to particular patterns: In particular, prices tend to react faster and stronger to market pressures than rents, typically leading to *decreasing* price-rent ratios during down-turns (see e.g., Geneseve and Mayer, 2001; Engsted and Pedersen 2015; Hill and Syed 2016; Francke and Korevaar 2021).

We compute list *price-rent* ratios using the "double-prediction" (also called "double-imputation") approach a hedonic method that is well established in the hedonic price index literature (see e.g., Hill and Syed 2016). Its main benefit is that it partially addresses the omitted variable problem, which can be particularly important when comparing properties from two different data sets. We employ it for the price-rent ratio calculation because we are concerned that properties in the sales market systematically outperform those in the rental market on characteristics that are not contained in our data sets; for example, sales properties that are better maintained or have higher quality fittings. We do not expect such systematic biases to be present *within* each data set (e.g., when going from one year to the next), which is why we use the more straightforward Time Dummy hedonic method to establish hedonic indices when only one data set is concerned.

List volume, *list time-on-market*, and *list turnover rate*, are three measures of market liquidity. Liquidity in the housing market tends to be positively correlated with price, a phenomenon that is explained in the literature by "loss aversion" – the reluctance of sellers to sell at (nominal) losses (see Genesove and Mayer 2001) – and/or equity constraints – individuals being ''locked" into a mortgage unable to move (Stein 1995). Each indicator captures a separate aspect of housing market liquidity: *List volume* measures the number of unique properties on the market at a particular point in time or during a certain time period; it is a measure of supply and thus of the liquidity going *into* the market. *Time-on-market* measures how fast properties are selling and thus captures the liquidity *leaving* the market. *Time-on-market* tends to be negatively related to house prices, and has long been used as a measure of

liquidity in the real-estate market (Miller 1978; Lippman and McCall 1986; Genesove and Mayer 2001, 1997). The market *turnover rate* is another popular measure of housing market liquidity (Genesove and Mayer 2001; De Wit et al. 2013; Van Dijk and Francke 2018). An alternative name for the turnover rate in the literature is rate-of-sales. It combines volume and *time-on-market* aspects of liquidity, but loses some detail while doing so.

When scraping data from online property platforms in regular intervals (e.g., daily or weekly), the history of properties on the market can be tracked. We use these individual histories to construct a new type of *sentiment indicator* based on the number and direction of price changes at the individual level. This indicator is new to the literature, and we believe it is a useful addition to monitor the "nervousness" of sellers and landlords in the housing market. Overall, the market sentiment analysis confirms that sellers are more likely to lower prices the longer properties remain on the listing platform. However, the rate at which this happens differs significantly between London and Vienna. London sellers and landlords frequently change prices while their counterparts in Vienna rarely do so.

We construct all list-market indicators on a monthly basis for the London and Vienna market throughout 2019 and 2020. While indicators mostly agree in their assessment of the market, sometimes they point in different directions. The findings of this manuscript therefore reiterate the importance of analysing multiple indicators before judging the behaviour of the housing market based on data from online platforms.

3. Data

For list data, we scrape from the leading platform in each market: for Vienna this is Willhaben (2021), for London it is Zoopla (2021). We started to collect sales as well as rental offers for both cities in the autumn of 2018 and include data-updates until December 14, 2020. We focus on apartment transactions rather than houses as the majority of transactions in both markets are for apartments. In 2019, 82% of residential transactions where for apartments in Vienna, in London the proportion was 59%.

When constructing the price indices and price-rent-ratios we want to get as "close" as possible to true market transaction prices. To do this, we track each property based on its url-address and include only the last price of a property before it leaves the platform. Thus, each property in the data set is included only once in the price index for the entire two-year time period. See Table 1 for the relationship between total entries and last entries. Other indicators need the first as well as the last entry on the platform (e.g., list *time-on-market*), or even the entire pricing history (e.g., *market sentiment indicator*).

In terms of data cleaning, we proceed as follows: For London we remove all listings with more than seven bedrooms or five bathrooms. In the Vienna data set – where we have information about size in square meters – we include all apartments between 20- and 180 square meters. For the list-price index calculations, we keep only the last entry before a property leaves the platform; then, we exclude the biggest price outliers by removing the top and bottom 3% of the price distribution in the London data set, and the top and bottom 3% of the price/square meter distribution in the Vienna data set. We follow the same steps for the transaction data sets, where the only data entry of a property is automatically the last entry.

Table 1 provides an overview of our list-price data set before and after the cleaning stage. More details on the data and cleaning stages can be found in Appendix (Subsection

Table 1. Descriptive statistics of data sets (2018–2020).

City		Apartments for sale			Apartments for rent		
		Raw data set		Cleaned data set	Raw data set		Cleaned data set
		Total	Unique		Total	Unique	
London	Observations	4,119,119	335,506	167,391	4,233,084	939,416	589,199
	Median price (Pounds)	590,000	525,000	495,000	1,820	1,733	1,772
	CV	2.39	2.29	0.56	1.58	1.39	0.43
Vienna	Observations	434,69	163,640	137,650	347,935	177,280	151,176
	Median price (EUR)	345,000	320,000	299,990	949.48	899	898
	CV	1.40	1.31	0.53	1.05	1.12	0.42

Note: We scraped the London market daily, while the Vienna market was scraped only once per week.
Note: To allow a comparison between the price distributions with different units of measure, we also include the Coefficient of Variation (CV).

6.2.). Note that, as apartments are typically advertised for weeks or months before being rented or sold, the number of total listings is much higher than the number of unique listings in both rental and sales markets.

In the hedonic regressions, to provide a true comparison between markets, we included only those characteristics that had relatively few missing entries. Also, for each city, we limited ourselves to those characteristics that were available for both the sales as well as the rental market. In particular, we included the following characteristics:

- inside space: the number of bedrooms (1, 2, 3, 4+) and bathrooms (1, 2, 3, 4+) in London and square meters in Vienna.
- outside space (garden or terrace) for Vienna
- historical building yes/no for Vienna
- distance to the city centre (which we constructed from individual GIS locations)
- type of property (flat/maisonette for London and flat/penthouse for Vienna)
- region dummies

We construct our own region dummies. To achieve this, we divide the city area into nine separate areas by forming two concentric circles around the city center (defined as Trafalgar's Square in London and Stephansplatz in Vienna) and dividing them by the North/South East/West axis. Figure 1a and Figure 1b illustrate the resulting regional dummies. Note, that the innermost circle is considered as one area (not divided into four) and provides the base area in the hedonic regressions (see Tables 5 and 6).

As London and Vienna are of similar size with respect to the area enclosed within their city limits, we use the same calculation for both cities, however London's property density in the outer regions is much higher than that of Vienna.

In addition to these characteristics, the frequent scraping of data allows us to keep track of each individual apartment throughout its time on the market. Individual identifiers like url-addresses allow us to establish a personal (price) history of each property with respect

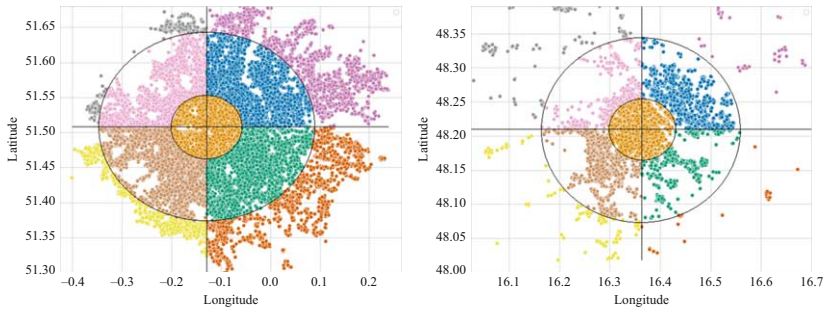


Fig. 1. Construction of regional dummies for London and Vienna.

to time on the market and/or price movements. This is definitely a benefit of (frequently scraped) list data, and we will use the pricing history to derive our sentiment indicator in Subsection 4.6.

We have access to a sample of transaction data for Vienna via the Austrian data provider ZT datenforum (2021). These data cover a 12 month period in 2019 during which all contracts are included that contain information on the interior space of the property (which is available for approximately 50% of transactions). The transaction data for London come from the UK Land Registry (2021a) and contain almost 100% of transacted properties. Table 2 shows a summary of the transaction data sets for the year 2019.

Figures 2 and 3 illustrate the relative geographic coverage of the list and transaction data sets. The spatial coverage of the list-price data overlaps with the transaction data, with the exception of a small part on the north-west side of London.

Figures 4 and 5 illustrate the distribution of prices before and after our cleaning process for list-price data compared with the corresponding transaction price data sets in 2019. Price negotiations between buyers and sellers, as well as inflated prices for properties that

Table 2. Summary statistics of transaction data sets, year 2019.

City	Observations	Median price	CV
London	56,601	415,000 Pounds	0.48
Vienna	5,755	258,864 EUR	0.59

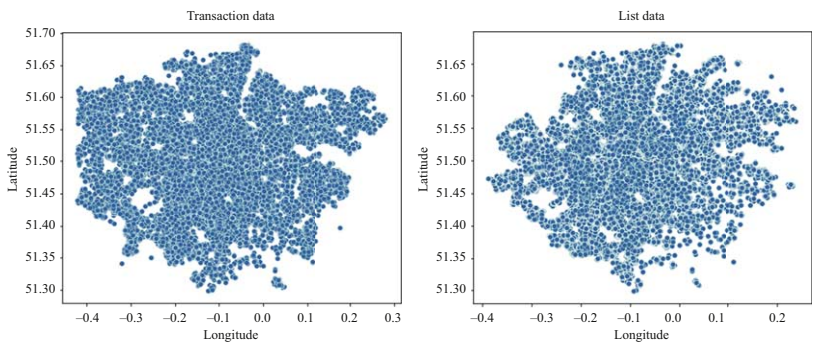


Fig. 2. Geographic coverage of list and transaction data London, 2019.

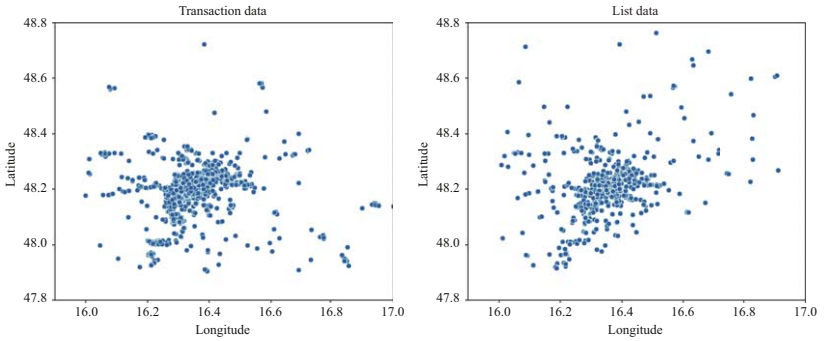


Fig. 3. Geographic coverage of list and transaction data Vienna, 2019.

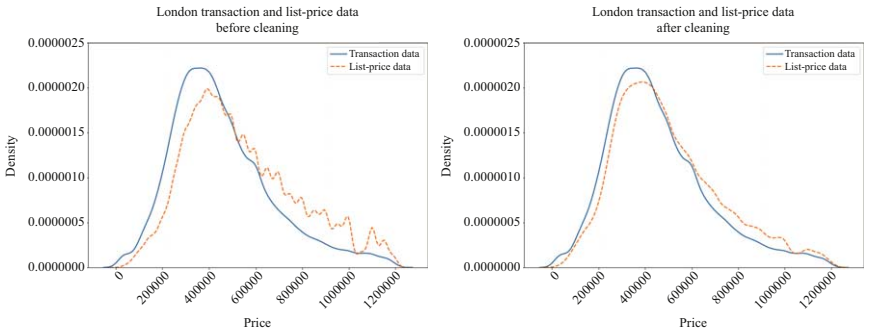


Fig. 4. London price data.

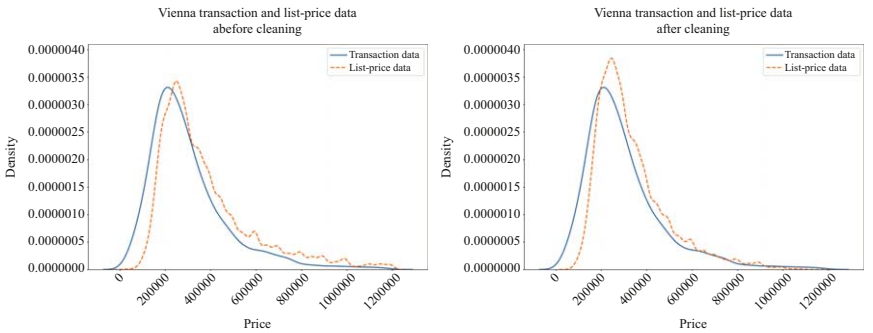


Fig. 5. Vienna price data.

are not being sold, will make list prices lie above transactions prices. Our cleaning process (especially selecting only the last price before a property leaves the platform) shifts the distribution of the list-price data towards that of transaction prices; this concerns in particular the highly-priced right tail of the listing data sets. With this process we find a closer correspondence between list and transaction data than others in the literature (see e.g., Shimizu 2016; Kolbe et al. 2021). Interestingly, on the left-hand side of the price distributions transaction prices lie above listing prices. This is probably due to non-market

transactions between family members, friends, or neighbours. The properties associated with such non-market transactions will show up in the transaction data set but be missing from the listing platforms (as there is no reason to advertise them).

4. Methods and Results

4.1. Indicator 1: List-price and List-rent Indicators

A house price index (HPI) measures changes in property prices across a designated market and hedonic indices are the standard tool for quality adjusting house price indices (see e.g., Eurostat 2013; IMF 2020). For some markets, sales or rent-determining characteristics are not available, and median price indices are used instead. We illustrate the difference between median and hedonic indices by looking at both types in this section.

Figure 6a depicts the percentage median price change per month for sales in London and Vienna. Figure 6b shows the median price index for rental properties in those cities. The most striking feature of Figure 6a is the dramatic increase in London median prices in April 2020 (by 14%) and the subsequent dramatic drop in the subsequent months. The period of the sudden increase in the median index coincides with the first physical COVID-19 shutdown period in London, which occurred from mid-March to mid-April 2020.

However, when inspecting the data closer, we find that this shutdown price spike is in a large part due to changes in market composition rather than actual price increases. During this first UK COVID-19 shutdown period properties still left the platform (presumably because they were sold) but hardly any new properties were added to the platform. Furthermore, the composition of properties on the London market changed during this period: Up-market properties were more likely to remain (and become) listed during the shutdown period than lower-priced properties. This resulted in the dramatic increase in the median apartment price in April, which is depicted in Figure 6a. Interestingly, the composition of listed apartments went almost back to "normal" in May and June 2020. This point illustrates how listings data could be utilised in parallel with transaction data as an early warning indicator of stock mix issues.

Hedonic indices are better able to control for compositional differences in the data than median price indices. We use the time-dummy method to construct hedonic list price index using data from November 2018 to December 2020 (for references on the time-dummy

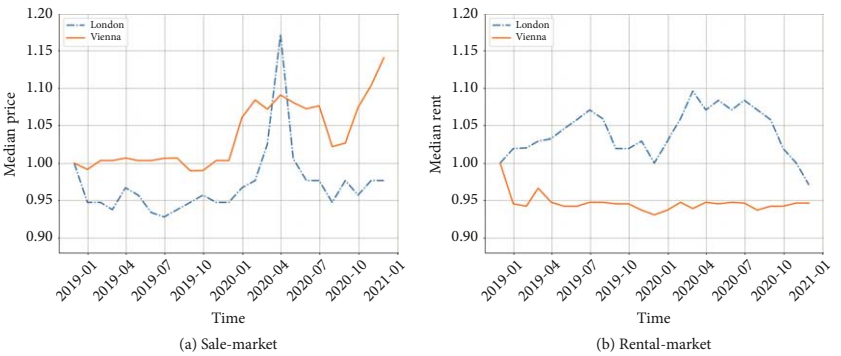


Fig. 6. Monthly median price development.

method, see e.g., Diewert et al. 2009; Eurostat 2013; Hill 2013). For each city we estimate separate semi-log hedonic sales as well as rental models as follows:

$$\ln(P^S) = X^S\beta^S + \gamma^S D^S + u^S \tag{1}$$

and

$$\ln(P^R) = X^R\beta^R + \gamma^R D^R + u^R. \tag{2}$$

The superscripts S and R in Equations 1 and 2 denote the sales and rental market, respectively. The matrices X^S and X^R contain all information on the characteristics for each property, the β terms indicate the corresponding estimated shadow prices, the γ are the estimated coefficients on the time dummy, and u^S and u^R are the error terms. $\ln(P^S)$ and $\ln(P^R)$ denote logged sales and rental list prices. The D matrices have the dimension H times T, where H is the total number of observations, and T corresponds to the number of time periods. Hence, each entry in our data set is linked to one row in the D matrices, where the elements for each row are filled with zeros, except for the column with the dates of the observed listings. The X matrices contain information on inside space, outside space (yes/no), new building (yes/no), and location (see Section 2 on the construction of region dummies).

The resulting list-price indices are derived by applying the exponent to the estimated time dummy coefficients γ and are illustrated in Figure 7. Both London and Vienna see rising prices in the sales market in the lead-up to the COVID-19 crisis, but price developments start to diverge in April 2020: the Vienna market continues its previous trend of strong price increases, while the London market softens.

The regression results for the London and Vienna markets are shown in Appendix (Section 6). The descriptive variables are highly significant for sales as well as rental markets and have the expected signs. Overall, the goodness of fit is better for the Vienna market (with an adjusted R^2s of 0.80 for the rental market and 0.77 for the sales market) than for London (with adjusted R^2 of 0.58 and 0.65 respectively). This is due to the better description of properties in Vienna, where we have information about the exact size of the property in square meters, whether it has an outside space, is located in a historic building, is in need of renovation, or can be considered a "penthouse". In contrast, in London, we

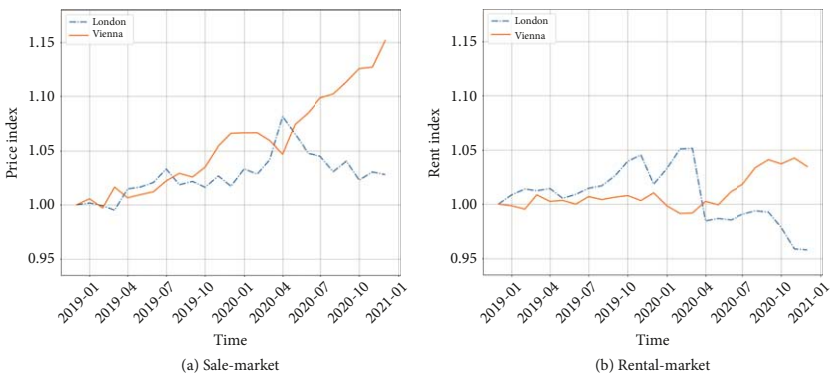


Fig. 7. Monthly list-price and list-rent indices for London and Vienna.

have much less detailed information on the individual properties, and size is measured in the form of bedroom and bathroom dummies rather than exact measurements. Distance to the city center has a negative impact in both cities, and the location dummies have the expected sign and are almost all highly significant.

Figure 7a illustrates the price movements for London and Vienna properties during our study period. Prices for apartments in Vienna increased during the entire period of observation (with the exception of a small dip in April during the first physical shut-down period). Compared to the median price index (see Figure 6a) the hedonic price index shows a much smoother price development. Comparing Figures 6a and 7a for London, the large price spike for April 2020 in the median price index is reduced to less than half of its previous size with the hedonic index. This example illustrates how important it is to control for quality in the construction of price indices, especially when working with online data that are not "tested by the market".

In terms of market direction since the beginning of the COVID-19 crisis in March 2020 (when the first lock-down periods were imposed in both the UK and Austria), the picture is clear in Vienna, where we find a rapidly increasing price index from April 2020 to December 2020. The interpretation of the London sales market is more difficult. After an initial price rise followed by a price fall during the first part of 2020, prices stabilized in the second half of the year. This could be (at least partially) due to a policy intervention by the UK government: it announced a sudden change in property taxation in July 2020, under which properties costing up to 500,000 pounds became excluded from property taxation between July 2020 and March 2021. Roughly half of the London transactions fall within this limit. The rental development throughout 2020 is very different in the two cities: rental prices in London decreased from April 2020 to December 2020, while Vienna rents increased during the same period.

For the rental markets, the differences between median and hedonic price indices are less dramatic, but still present (compare Figures 7b and 6b). Figure 7b illustrates the quality-adjusted rent-price indices for London and Vienna. Similar to the sales markets, the price developments for London and Vienna also diverge in the rental markets. While Figure 7b shows falling rents for London since the beginning of the COVID-19 crisis in February 2020, it indicates rising rents in Vienna for the same time period. Interestingly, this rent increase depicted by the hedonic rent index for Vienna is not visible in the median rent index; The median index cannot pick up that over the two-year period rental properties in Vienna became increasingly smaller.

We now turn to the question of whether list price indices are good predictors for transaction-based price indices. Statistics Austria computes transaction price indices for apartments in Austria but does not publish a separate price index for Vienna. The Austrian National Bank provides a separate price index for Vienna, but it is based mostly on list-price data. Thus, for Vienna, we cannot compare list and transaction price indices directly but will illustrate the close co-movement of our list-price index with the list-price index of the Austrian National Bank (OeNB, 2020). Since the OeNB index is provided on a quarterly basis, we also include a quarterly index for Vienna (see Figure 8).

For London we can compare list-price and transaction-price indices: In Figure 9 we plot the official ONS transaction price index for London together with our lagged hedonic *list-price index*. The official HPI for London (https://landregistry.data.gov.uk/app/ukhpi) is

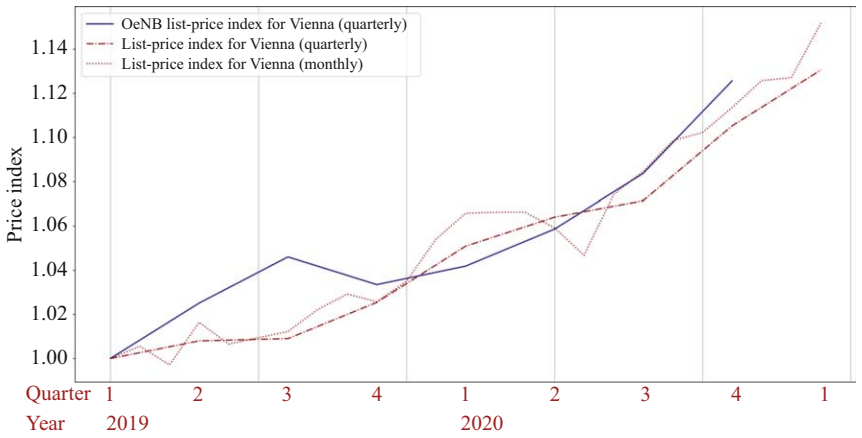


Fig. 8. Comparison of list price indices for Vienna.
Source: Own calculation; Austrian National Bank (OeNB).

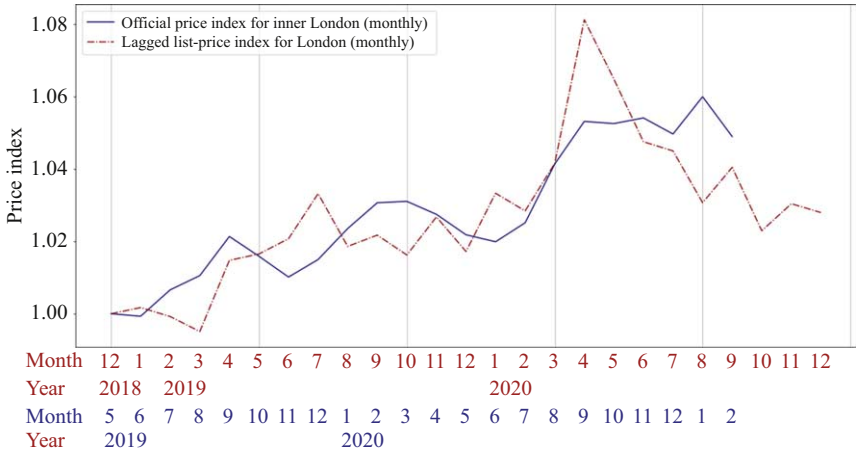


Fig. 9. Official house price index – London.
Note: There are two labels on the x-axis. The first row (red) corresponds to the list-price index that we lagged by five months. The blue labels correspond to the results for Inner London computed by the UK Land Registry. Own calculation; Source: UK Land Registry.

available at quarterly frequency for Inner London (this corresponds to our boundaries) as well as for Greater London, and the City of London. Last retrieval: 20.05.2020 together with our lagged hedonic list-price index (see UK Land Registry 2021b). As we do not have long enough time-series data to estimate the lag between prices appearing on the internet platform and transactions econometrically, we chose a five-month lag based on visual inspection. Our five-month lag fits well within the range of lags discussed in the literature. Longer lags of about seven months have been found for Tokyo (Shimizu et al. 2016), while Brounen and Kok (2011) find a shorter lag of four, and a half months for the Dutch housing market. An estimate for the UK market presented in Eurostat (2013, 103) indicates a five, and a half months lag.

The overall movements of the two indices coincide, even though the ONS index contains price movements of apartments and houses while our index concentrates on apartments only. The transaction price index is a bit more "jumpy" compared to the *list-price index*. This is probably due to the greater price stickiness of list prices compared to transaction prices – a phenomenon that has been discussed in the literature by Haurin et al. (2013). Note, that we cannot draw any conclusions about relative price levels from these figures as we normalized the listing and transaction price indices to 1 at the beginning of the period. Actual transaction data would be needed for a comparison of list- and transaction prices in levels.

4.2. Indicator 2: List Price-Rent Ratios

We compute *list price-rent ratios* using the "double-prediction" (also called "double-imputation") approach a hedonic method that is well established in the price index literature (see e.g., Hill and Syed 2016).

The method works as follows: First, we estimate two hedonic models, one for the sales market and one for the rental market as done in the previous section. Second, we use these models to impute a rental price for each sales object as well as a sales price for each rental object. Third, we compute "own-imputation": this involves imputing the price of each sales property via the hedonic sales model as well as the rental price for each rental property via the hedonic rental model. Fourth, we compute price-rent ratios for each property. Finally, these individual (imputed) price-rent ratios are aggregated to the market price-rent ratio.

Thus, for apartment j with characteristics c, which is listed for sale, we estimate:

$$\frac{\hat{P}^S(x_j^S)}{\hat{P}^R(x_j^S)} = \frac{exp\left(\left[\sum_{c=1}^{C}\hat{\beta}_c^S x_{j,c}^S\right] + \gamma^S d_j^S + \varepsilon_j^S\right)}{exp\left(\left[\sum_{c=1}^{C}\hat{\beta}_c^R x_{j,c}^S\right] + \gamma^R d_j^S + \varepsilon_j^R\right)}, \tag{3}$$

where $\ln \hat{P}^S(x_j^S)$ and $\ln \hat{P}^R(x_j^S)$ denote the predicted log offer price and log rental price for property j in the sales data set, and $\hat{\beta}^S$ and $\hat{\beta}^R$ are the estimated characteristic shadow prices obtained from Equations (1) and (2). γ^S and γ^R are the derived time coefficients, and ε^S and ε^R are the error terms for the sales and rental market respectively.

Similarly, for apartment h with characteristics c, which is offered for rent, we estimate the following:

$$\frac{\hat{P}^S(x_h^R)}{\hat{P}^R(x_h^R)} = \frac{exp\left(\left[\sum_{c=1}^{C}\hat{\beta}_c^S x_{h,c}^R\right] + \gamma^S d_h^R + \varepsilon_h^S\right)}{exp\left(\left[\sum_{c=1}^{C}\hat{\beta}_c^S x_{h,c}^R\right] + \gamma^R d_h^R + \varepsilon_h^R\right)}, \tag{4}$$

To derive at the overall price-rent ratio, we take the geometric mean as follows:

$$Med[\hat{P}^S/\hat{P}^R] = \sqrt{Med[\hat{P}^S/\hat{P}^R]_{sales} \times Med[\hat{P}^S/\hat{P}^R]_{rental}}, \tag{5}$$

where $Med[\hat{P}^S/\hat{P}^R]_{sales}$ and $Med[\hat{P}^S/\hat{P}^R]_{rental}$ represents the median of the price to rent ratio derived from Equations (3) and (4) respectively.

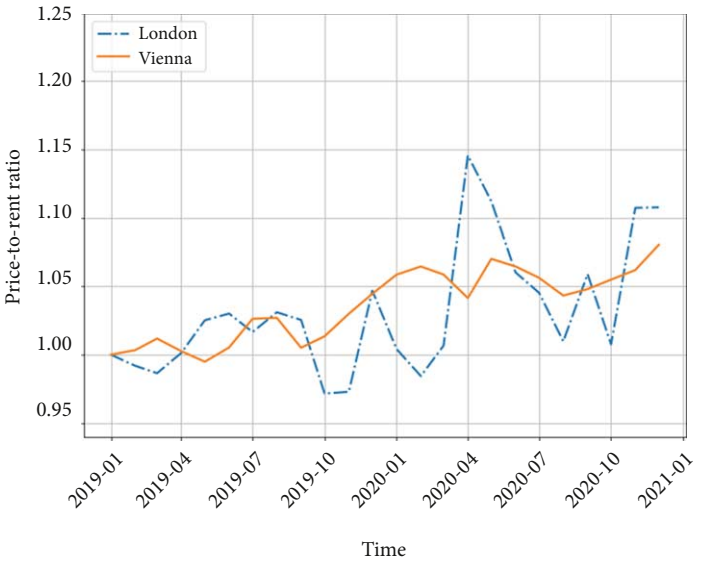


Fig. 10. Monthly list price-rent ratios.

The resulting *list price-rent* ratios for both cities are shown in Figure 10. In the Vienna market the *list price-rent ratio* increased in a smooth pattern over the entire two year period. Results for London again show an "April-peak" and a much more volatile up and down movement of the price-rent ratio – especially for 2020. As a result, it is unclear how the price-rent ratio should be interpreted for the London market.

4.3. Indicator 3: List Volume

Market volume, the number of properties transacted during a particular time period, is an important measure of market tendency. This is the case since housing markets tend to be more liquid during market up-turns than during down-turns. The higher liquidity during market up-turns implies that more transactions are taking place and that houses are selling more quickly. During market down-turns there are typically fewer sales and houses tend to remain on the market for longer. Different theories for this stylized fact of the housing market exist. Stein (1995) links the phenomenon to credit constraints, while Genesove and Mayer (2001) illustrate how loss aversion could be the reason sellers refuse to sell during market down-turns (at prices below the original purchase price).

List volume is a related, but different, aspect of market liquidity and measures the number of unique properties on the market at a particular point in time. To construct this indicator we compute the number of unique listings per month for both the sales as well as the rental market. Listings that are available for several months will enter the volume count once each month (duplicates within one month are ignored). To compare "like with like" we include the same number of days for each month by shortening the longer months. Furthermore, we normalize the series to the number of unique properties listed in December 2018 in each market. It is important to note here that we can only measure short-term housing supply with this indicator.

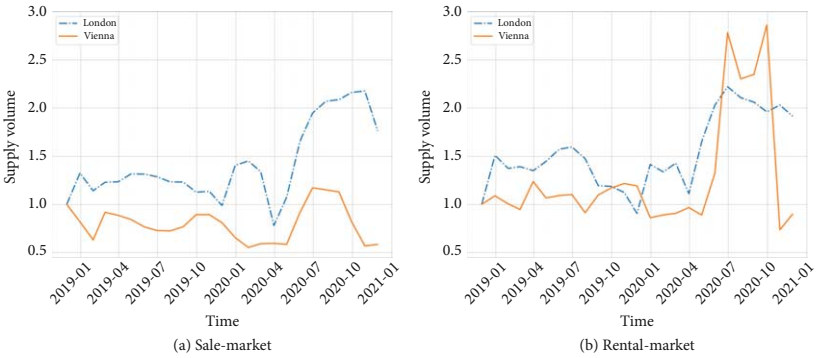


Fig. 11. Total number of listings per month.

Figure 11a illustrates the *list volume* in the sales market for both cities. As already discussed above, there is a sharp decrease in listed properties in the London market during March/April 2020, followed by a strong increase in subsequent months. On the online platform the number of unique properties for sale in London dropped from over 10,000 per month in February 2020 to less than 6,000 in April, only to rise to over 14,000 by the end of July 2020. No such drop in *list volume* is visible for the Vienna market but *list volume* is more volatile in 2020 than in 2019.

List volume for rental properties is shown in Figure 11b. The number of listed rental apartments in London doubled from April to June 2020 and stayed at this high level for the rest of the study period. The sudden increase in available rental properties after April 2020 was even stronger in Vienna. Suggestions in the media speculate about an "Airbnb effect" that led to the sudden jump in available rental properties (Fonds Online 2020). This theory is plausible for Vienna as these new rentals are located to a large degree in the touristic inner districts of Vienna. The number of listed properties stayed high until a sudden drop in November 2020.

4.4. Indicator 4: List Time-on-market

A large benefit of daily list price data is that *time-on-market* can be measured at the level of individual properties. The aggregate *time-on-market* indicator here is defined as the mean number of days properties have been advertised on the platform. However, as not all properties that are taken off the market are also sold, the *list time-on-market* indicator will generate a lower bound for actual *time-on-market*. Figure 12 below illustrates this concept. The importance of relisted properties in defining and assessing *time-on-market* is investigated in Benefield and Hardin (2015) who also note that earlier estimates of *time-on-market* based on printed adverts in newspapers are probably even more biased downwards than calculations based on listing platforms.

Listing starts have to be approximated for the Vienna data set as first appearance on the website is not one of the available characteristics. This means that *time-on-market* will steadily increase in the initial months before settling into its true pattern. We highlighted this initial period with gray in Figure 13a and 13b. For the London market, we are able to derive the *time-on-market* measure based on the provided information of each first listing

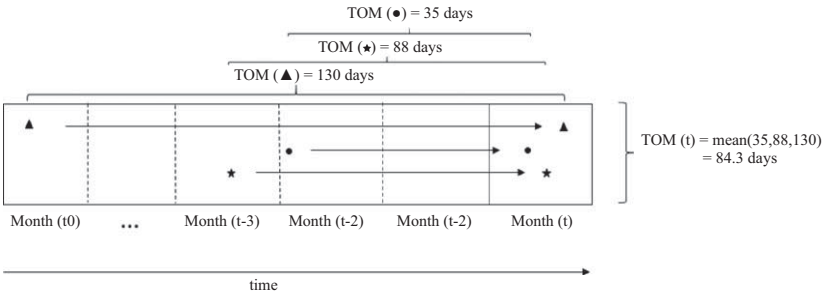


Fig. 12. Illustration of time-on-market calculation.
Note: TOM refers to *time-on-market*.

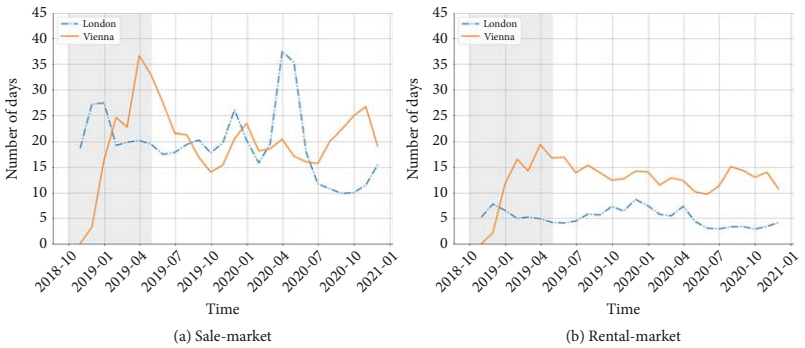


Fig. 13. Mean time-on-market.

entry from the website provider. Hence, the *time-on-market* can be calculated from the beginning (i.e., without lead time).

Time-on-market for rental apartments are also computed in the same way and are shown in Figure 13b. As expected, rental apartments have a shorter *time-on-market* than apartments for sale.

After the onset of the COVID-19 crisis, the *time-on-market* for apartments for sale in London exhibits a sharp up and then down-turn: while it sharply increased from February to May, there was a sharp trend-reversal in the summer of 2020. This drop in *time-on-market* coincided with the suddenly imposed stamp duty holiday for properties under 500,000 pounds for England and Northern Ireland, which started in July 2020 and lasted until March 2021. Roughly half of the apartments in London lie below this 500,000 pound threshold (see Table 1) and are thus exempt from stamp duty during this period (before July 2020 the cut-off level for stamp duty exemption was 125,000 pounds). In contrast to the decreasing *time-on-market* for sales properties in London, we see a smooth and stable *time-on-market* in the London rental market for the entire period of observation.

In Vienna, *time-on-market* for both the sales and rental market was relatively stable throughout 2020. The Vienna sales market even saw a decline in *time-on-market* for the first six months after the onset of the COVID-19 crisis. A similar slight downward trend in *time-on-market* is also noticeable for the Vienna rental market until mid-2020.

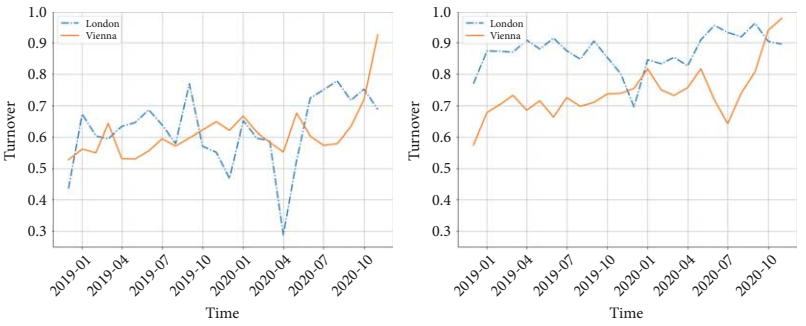
4.5. Indicator 5: List Turnover Rate

The *list turnover rate* is constructed using a similar methodology to the classic market turnover rate, which is generally defined as the number of sales (transactions) in a period divided by the number of properties available at the beginning of the period. The market turnover rate is a frequently discussed measure of market liquidity which (like market volume) tends to be positively related to price (see Genesove and Mayer 2001; De Wit et al. 2013; Van Dijk and Francke 2018). For theoretical explanations of this positive relationship see discussions on (nominal) loss aversion (e.g. Genesove and Mayer 2001), and equity constraints (e.g. Stein 1995; Genesove and Mayer 2001).

Here, the list *turnover rate* is constructed by dividing the number of properties that leave the platform (i.e., are available at some time during the month but disappear before the end of the month) by the total number of (unique) properties available on the platform during that month. The results for the *turnover rate* are shown in Figures 14a and 14b.

The *turnover rate* lies around 0.6 for both the London and Vienna sales markets until the start of the COVID-19 crisis in March 2020. A turn-over rate of 0.6 implies that 60% of properties that are available at some time during the month are no longer available at the end of the month. Interestingly, the turnover rates for the two cities then diverged dramatically in the spring of 2020. Turnover dropped sharply in London between February and April 2020 and only 27% of the properties on the platform in April 2020 also left the platform during that month. However, this turnover rate rebounded sharply and two months later almost 80% of properties left the platform within one month. The higher *list turnover rate* in the London sales market in the second half of 2020 is probably related to the stamp duty holiday, which was put in place in July (and runs until March 2021). The *list turnover rate* in the sales market in Vienna also becomes more volatile after the start of the COVID-19 crisis, but at a higher level than in London. Overall the *list turnover rate* in Vienna increased after the start of the COVID-19 crisis, another indication of a heating property market.

In line with the lower *time-on-market*, *list turnover rates* in rental markets are higher than in sales markets and for both markets there seems to be an upward trend in 2020. The fall in turnover in the Vienna rental market during the summer of 2020 could be influenced by Airbnb landlords shifting short-term rental properties to the longer-term rental market.



(a) Sale-market (b) Rental-market

Fig. 14. Turnover Rate

4.6. Indicator 6: Sentiment Indicator Changes

A great advantage of daily scraped list price data is that it has a history: Each property can be identified via its unique url-address which allows us to observe not only its current offer price but also its entire pricing history. Here we use this pricing history (in particular the direction and frequency of price changes) to construct an indicator that informs about sellers' market sentiments. Properties tend to have more negative than positive price changes during their stay in the market as sellers often "try out" higher prices first before decreasing them later. However, disturbances to this "normal" pattern (i.e., more positive or negative price changes than usual) can be used as an indicator for shifting market sentiment. As markets differ with respect to what is considered a normal pricing strategy, an indicator based on such price changes will be specific to each market.

We derive our market sentiment indicator as follows:

1. At the end of each month, we establish the direction of the last price change for each apartment,
2. We construct a variable that reports "+ 1" if the last change was positive, "-1" if the last change was negative, and "0" if the price remained unchanged since the apartment entered the market. Figure 15 illustrates the process, and
3. We then take the sum of this variable over all properties and divide it by the total number of apartments for sale during the month.

Results for this variable are shown in Figures 16a and 16b. The most striking finding is that overall pricing strategies differ dramatically between London and Vienna. While London sellers tend to lower prices during a property's stay on the platform, sellers in the Vienna market hardly ever change prices – and if they do, they are almost as likely to increase as to decrease them. To further illustrate this difference we include Table 3, which

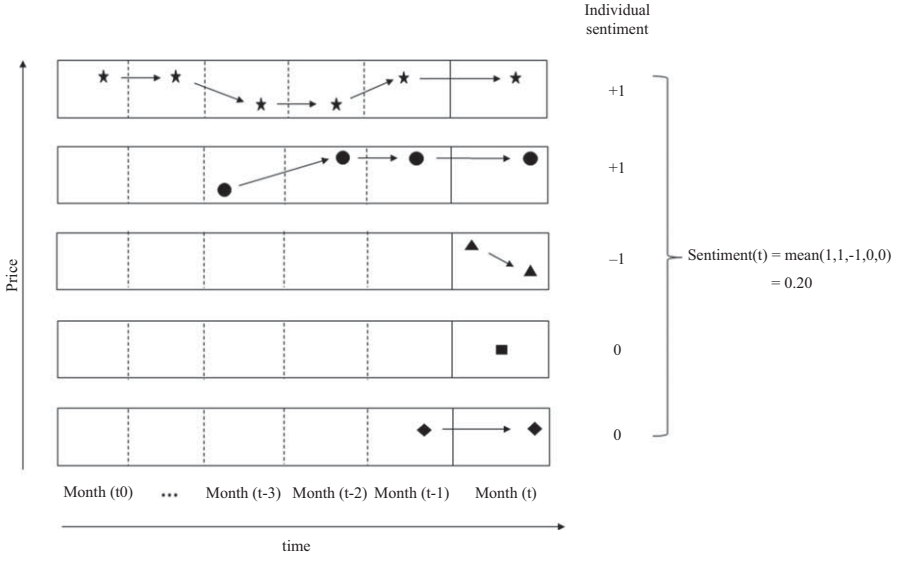
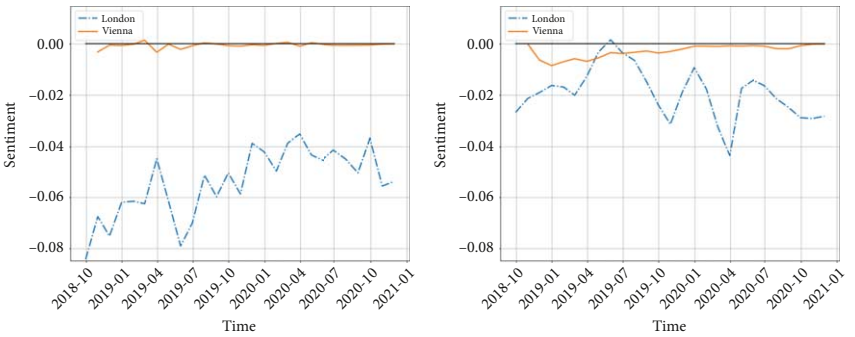


Fig. 15. Illustration of sentiment calculation.



(a) Sales-market (b) Rental-market

Fig. 16. Market sentiment per month.

shows the total number as well as the composition of monthly price changes in all submarkets. It clearly illustrates that more changes are made by London property holders than by their Vienna counterparts.

We turn now to the history of the sentiment indicators over the two-year period. Even though the sentiment indicators for the sales markets in London and Vienna differ in levels, Figure 16a shows that sellers in both markets were less likely to lower prices in 2020 than 2019. This result fits well with the positive (even though, in the case of London, somewhat volatile) price developments shown by the hedonic price indices in Subsection 4.1.

Figure 16b illustrates the sentiment indicators for the rental markets. Similarly to the sales market, London landlords are more likely to lower prices than their counterparts in Vienna. It is notable that the London landlords' sentiment was lower in 2020 than in 2019. Thus, the sentiment of London landlords and London sellers in 2020 did not match according to this indicator. In contrast, Viennese landlords were less likely to lower rents for advertised apartments in 2020 than in 2019, and the sentiment in the rental market shows a prolonged upward trend since the beginning of 2019.

4.7. Summary of Indicator Results Throughout 2020

Table 4 provides a summary overview of the results discussed in Subsections 4.1 to 4.6. Here we focus on the broad direction of the indicators from the start of the COVID-19 crisis until December 2020. In Table 4, we take the average indicator values for January, February, and March 2020 as the pre-crisis reference values to compare the December 2020 values with. To test the robustness of these results we also investigated two alternative definitions of the pre-crisis period; the direction of the arrows in Table 4 remain unchanged (see C for more details).

The overall liquidity measure in Table 4 is based on the combined movement of the three liquidity measures – *volume*, *time-on-market*, and *list turnover rate*. Each month we measure the tendency for each of these three liquidity measures; the overall monthly liquidity measure is positive if two of the three liquidity measures indicate a positive liquidity tendency. Table 14 in the Appendix illustrates the movements for each of these liquidity measures as well as their combined direction each month.

Table 3. Number of positive (+) and negative (−) price changes per month.

Time	London						Vienna					
	Sale		Total change	Rent		Total change	Sale		Total change	Rent		Total change
	(+)	(−)		(+)	(−)		(+)	(−)		(+)	(−)	
10-2018	793	6,746	10.69%	1,214	2,984	6.38%	–	–		–	–	
11-2018	1,609	10,177	9.30%	2,440	5,584	5.46%	0	7	0.31%	0	0	0.00%
12-2018	865	6,766	9.73%	1,819	4,044	5.03%	18	23	0.31%	5	56	0.73%
01-2019	1,261	6,766	9.04%	2,002	3,923	5.03%	74	85	0.85%	79	199	1.97%
02-2019	1,215	5,062	10.04%	1,531	3,051	5.09%	152	161	1.38%	110	245	1.81%
03-2019	830	4,549	9.05%	1,342	2,827	5.65%	163	126	0.91%	46	196	0.95%
04-2019	953	3,889	7.43%	1,643	2,693	5.35%	82	205	0.76%	63	291	1.06%
05-2019	957	5,359	8.85%	1,968	2,230	4.86%	202	213	0.94%	49	263	0.80%
06-2019	757	5,639	10.37%	2,488	2,349	5.64%	72	176	0.52%	45	193	0.54%
07-2019	888	5,645	9.63%	1,852	2,157	4.50%	151	189	0.64%	55	245	0.59%
08-2019	941	4,960	7.54%	2,577	3,229	5.56%	130	106	0.40%	60	247	0.54%
09-2019	1,136	4,718	9.77%	2,197	3,349	7.13%	128	123	0.40%	53	228	0.45%
10-2019	1,152	5,412	7.77%	2,271	4,414	7.44%	110	157	0.37%	63	315	0.53%
11-2019	1,080	5,901	8.50%	1,995	4,620	7.96%	110	181	0.37%	57	291	0.44%
12-2019	1,424	4,908	7.05%	2,738	4,977	6.43%	135	160	0.32%	77	259	0.37%
01-2020	1,693	6,236	7.37%	3,526	4,806	6.03%	151	205	0.36%	91	191	0.24%
02-2020	1,979	8,202	8.14%	3,008	5,299	6.41%	120	90	0.16%	57	181	0.10%
03-2020	2,999	10,133	7.13%	3,300	8,637	7.26%	277	197	0.36%	93	279	0.22%
04-2020	703	2,191	6.87%	2,466	8,316	8.02%	94	218	0.25%	40	144	0.12%
05-2020	573	2.267	7.28%	1,349	2,692	5.23%	160	155	0.22%	69	238	0.19%
06-2020	794	3,684	7.06%	1,258	2,461	4.37%	55	99	0.11%	24	12	0.09%
07-2020	759	3,869	6.17%	1,194	2,700	4.22%	78	148	0.18%	65	178	0.18%
08-2020	688	3,785	6.50%	1,222	3,101	4.94%	110	179	0.22%	102	367	0.33%
09-2020	720	4,462	6.98%	1,057	1,158	5.00%	144	212	0.25%	64	364	0.27%
10-2020	851	3,901	5.75%	1,014	3,732	5.06%	33	94	0.09%	18	128	0.09%
11-2020	553	3,539	7.62%	815	2,738	5.41%	20	35	0.04%	1	19	0.01%
12-2020	168	1,136	7.25%	517	1,752	5.21%	5	6	0.01%	1	6	0.00%

Table 4. Overall market development for London and Vienna for 2020.

		Price development	Price-to-rent ratio	Liquidity measure	Market sentiment
London	Sales	→			→
	Rental	↘	↗	↗	
Vienna	Sales			→	→
	Rental	↗	↗	↗	↗

5. Conclusion

In this article we used data from online real-estate platforms to investigate the initial impact of the COVID-19 crisis on the London and Vienna real-estate markets in 2020. Based on our data set, which we scraped from leading housing platforms throughout 2019 and 2020, we constructed early market indicators for *list-price*, *list-rent*, *price-rent ratio*, *market volume*, *time-on-market*, *market turnover*, and *market sentiment*. The last of these, market sentiment, is new to the literature. Even though the other indicators have been discussed before, we add to the literature by presenting them all within one article and by illustrating how to construct them using online price data. The examination of trends in the real estate market is particularly important during times of economic upheaval, and these online indicators allow us to do so before transaction data become available.

We took great care in preparing our online data set and constructing the indicators. In particular, our method of selecting only the last price before a property leaves the platform shifts the distribution of the list-price data towards that of transaction prices. This is particularly noticeable at the right tail of the listing data set. We believe that this closeness in price distributions illustrates that the list price-indices developed here can help predict future market developments. For the period of (lagged) overlap between list and transaction data, the official transaction price indices follow our list-price indices with a five-month lag. For Vienna, the *list-price index* anticipated the strong positive price development and, in London, the stable price level throughout 2020.

For the construction of the *list price-rent ratios*, we used the hedonic "double-prediction" approach in order to limit the influence of unobserved differences between the online sales and online rental properties. The *list price-rent ratios* increase throughout 2020 in both markets, driven by the price increases in both markets, and – in London – also by falling rents.

To measure housing volume, we looked at three different measures: *list volume*, *list time-on-market*, and *list turnover rate*. Each of these measures captures a slightly different aspect of market volume. When all three agree in their market assessment, they send a clear signal about current market tendencies. For the 2020 COVID-months this signal indicated increased liquidity in both markets.

The *sentiment indicator*, which we constructed from information on the direction and frequency of online price changes, is new to the literature. It aims to capture current sellers' market perception. According to this indicator, the sentiment of London sellers was relatively stable throughout 2020 and higher than during 2019. London landlords on the other hand were more positive in 2019 than in 2020. In addition to being more negative

than the year before, the sentiment indicator for the 2020 London rental market is also very volatile, exhibiting a big drop in measured sentiment during the first COVID-19 lockdown a sharp recovery in the summer, and a subsequent slow decline until the end of the year. In Vienna, both sales and rental market sentiments were buoyant during 2020.

Even with six different indicators it is hard to judge the overall state of the London sales market during much 2020. We see a sudden rise and then fall in sales prices after the first COVID-19 lockdown in March/April 2020. Then prices stabilize in the summer. A similar volatile pattern is present in the *list price-rent ratio*, the sentiment indicator, as well as the three market liquidity indicators. In contrast to this behaviour in the sales market, the direction of the rental market is easy to interpret: it became negative as soon as the crisis hit.

In contrast to the somewhat contradictory signals that we find for the London market, the Vienna housing market shows a strong upward trend over the entire two year period we investigated, and this trend strengthened after the start of the COVID-19 crisis in March 2020. In Vienna, both price and rent indicators continued their previous upwards trends throughout 2020, and time-on-market fell dramatically after the start of the pandemic. Underlining this market tendency is a price-rent ratio that is also moving upwards. Liquidity indicators have been more volatile in Vienna after the start of the COVID-19 crisis, but at a higher level than in London. In terms of historical perspective, the Vienna property market is no stranger to rallying in the wake of a global crisis: It also did so during the Great Financial Crisis between 2007 and 2010 when Vienna's prices rallied while other real estate markets fell (OeNB 2019).

We believe we illustrated with this article that list-data indicators can provide useful early warning signals for the development of the housing market. While we advise against interpreting list-data indicators as perfect substitutes for their transaction-based counterparts – they primarily illustrate the behaviour of the supply side of the market they are a rich data source that can be used to get market insights. In particular, constructing a number of these list-data indices and interpreting their joint behaviour can provide useful and timely information on future market movements.

6. Appendix

6.1. Model Statistics

London

Table 5. Model summary for London list-price index.

	Goodness of fit: London	
	Rental model	Sale model
Constant	7.655***	13.449***
	(0.002)	(0.006)
2019_1	0.009555	0.002
	(0.003)	(0.006)
2019_2	0.014***	−0.001
	(0.003)	(0.007)
2019_3	0.012***	−0.005
	(0.003)	(0.007)

Table 5. Continued

	Goodness of fit: London	
	Rental model	Sale model
2019_4	0.014***	0.016**
	(0.003)	(0.007)
2019_5	0.005*	0.016**
	(0.003)	(0.006)
2019_6	0.009***	0.021***
	(0.003)	(0.006)
2019_7	0.014***	0.033***
	(0.003)	(0.006)
2019_8	0.017***	0.018***
	(0.003)	(0.007)
2019_9	0.026***	0.021***
	(0.003)	(0.006)
2019_10	0.039***	0.016**
	(0.003)	(0.007)
2019_11	0.044***	0.026***
	(0.003)	(0.007)
2019_12	0.018***	0.017**
	(0.003)	(0.007)
2029_1	0.032***	0.033***
	(0.003)	(0.006)
2020_2	0.050***	0.028***
	(0.003)	(0.006)
2020_3	0.050***	0.041***
	(0.003)	(0.007)
2020_4	−0.016***	0.078***
	(0.003)	(0.009)
2020_5	−0.013***	0.063***
	(0.003)	(0.007)
2020_6	−0.015***	0.046***
	(0.003)	(0.006)
2020_7	−0.010***	0.044***
	(0.003)	(0.006)
2020_8	−0.006**	0.030***
	(0.003)	(0.006)
2020_9	−0.007***	0.040***
	(0.003)	(0.006)
2020_10	−0.022***	0.023***
	(0.003)	(0.006)
2020_11	−0.042***	0.030***
	(0.003)	(0.006)
2020_12	−0.043***	0.028***
	(0.003)	(0.006)
bathroom_2.0	0.251***	0.327***
	(0.001)	(0.002)
bathroom_3.0	0.453***	0.623***
	(0.003)	(0.006)
bathroom_4.0	0.414***	0.682***
	(0.010)	(0.029)
bedroom_2.0	0.180***	0.173***

Table 5. Continued

	Goodness of fit: London	
	Rental model	Sale model
	(0.001)	(0.002)
bedroom_3.0	0.327***	0.308***
	(0.001)	(0.003)
bedroom_4.0	0.371***	0.347***
	(0.002)	(0.006)
Distance to centre	−0.041***	−0.056***
	(0.000)	(0.000)
Maisonette	−0.022***	−0.040***
	(0.002)	(0.003)
$>5km$ from centre ne	−0.135***	−0.213***
	(0.001)	(0.003)
$>5km$ from centre nw	−0.048***	−0.017***
	(0.001)	(0.003)
$>5km$ from centre se	−0.120***	−0.173***
	(0.001)	(0.003)
$>5km$ from centre sw	−0.026***	−0.007**
	(0.001)	(0.003)
$>15km$ from centre ne	0.005	−0.045***
	(0.003)	(0.006)
$>15km$ from centre nw	0.021***	0.116***
	(0.005)	(0.009)
$>15km$ from centre se	−0.061***	−0.052***
	(0.004)	(0.006)
$>15km$ from centre sw	0.003	−0.047***
	(0.003)	(0.006)
Observations	551,413	159,570
adjusted R^2	0.575	0.650
Residual std. error	0.253 (df = 551372)	0.286 (df = 159529)
F statistic	18681.900*** (df = 40.0; 551372.0)	7410.401*** (df = 40.0; 159529.0)

Note: *p < 0.1; **p < 0.05; ***p < 0.01

Vienna

Table 6. Model summary for Vienna list-price index.

	Goodness of fit: Vienna	
	Rental model	Sale model
Constant	5.985***	11.719***
	(0.003)	(0.003)
2019_1	−0.002	0.006
	(0.003)	(0.004)
2019_2	−0.005	−0.003
	(0.003)	(0.004)
2019_3	0.008**	0.016***
	(0.003)	(0.004)
2019_4	0.003	0.007
	(0.003)	(0.004)
2019_5	0.003	0.009**
	(0.003)	(0.004)
2019_6	−0.000	0.012***
	(0.003)	(0.004)
2019_7	0.007**	0.022***
	(0.003)	(0.004)
2019_8	0.004	0.029***
	(0.003)	(0.004)
2019_9	0.006**	0.025***
	(0.003)	(0.004)
2019_10	0.008**	0.034***
	(0.003)	(0.004)
2019_11	0.003	0.053***
	(0.003)	(0.004)
2019_12	0.010***	0.064***
	(0.003)	(0.004)
2020_1	−0.002	0.064***
	(0.003)	(0.004)
2020_2	−0.009**	0.064***
	(0.004)	(0.005)
2020_3	−0.008**	0.058***
	(0.003)	(0.004)
2020_4	0.003	0.046***
	(0.003)	(0.005)
2020_5	−0.001	0.072***
	(0.003)	(0.004)
2020_6	0.012***	0.081***
	(0.003)	(0.004)
2020_7	0.018***	0.094***
	(0.003)	(0.004)
2020_8	0.033***	0.097***
	(0.003)	(0.004)
2020_9	0.040***	0.107***
	(0.003)	(0.004)
2020_10	0.036***	0.118***

Table 6. Continued

	Goodness of fit: Vienna	
	Rental model	Sale model
	(0.003)	(0.004)
2020_11	0.042***	0.120***
	(0.004)	(0.005)
2020_12	0.034***	0.141***
	(0.005)	(0.006)
Historic building	0.094***	0.068***
	(0.001)	(0.001)
Penthouse	0.107***	0.056***
	(0.012)	(0.008)
Need of renovation	−0.248***	−0.238***
	(0.017)	(0.005)
Outspace	0.073***	0.078***
	(0.001)	(0.002)
Square meter	0.012***	0.014***
	(0.000)	(0.000)
Distance to centre	−0.010***	−0.015***
	(0.000)	(0.000)
$> 5km$ from centre ne	0.028***	−0.053***
	(0.002)	(0.002)
$> 5km$ from centre nw	−0.006**	0.052***
	(0.003)	(0.004)
$> 5km$ from centre se	−0.019***	−0.179***
	(0.002)	(0.003)
$> 5km$ from centre sw	−0.011***	−0.026***
	(0.002)	(0.002)
$> 15km$ from centre ne	0.031***	0.039***
	(0.008)	(0.013)
$> 15km$ from centre nw	−0.006	−0.067***
	(0.005)	(0.008)
$> 15km$ from centre se	−0.059***	−0.259***
	(0.009)	(0.015)
$> 15km$ from centre sw	−0.001	0.013*
	(0.004)	(0.007)
Observations	151,176	137,650
adjusted R^2	0.802	0.770
Residual std. error	0.164 (df = 151137)	0.222 (df = 137611)
F Statistic	16134.656*** (df = 38.0; 151137.0)	12099.229*** (df = 38.0; 137611.0)

Note: *p < 0.1; **p < 0.05; ***p < 0.01

6.2. Descriptive Statistics of Missing Characteristics

Table 7. Data cleaning steps – London.

	Rental		Sale	
	Observations	Median price	Observations	Median price
Total unique	939,466	1,733	335,506	525,000
Property type not flat or maisonette	232,903	1,549	129,452	600,000
missing bedrooms and/or missing bathrooms	78,441	1,842	27,702	475,000
bedrooms >7 and/or bathrooms >5	250	15,167	83	7,600,000
price $>Q_{97}$	19,020	7,800	5,523	3,200,000
price $<Q_3$	19,653	693	5,355	137,500
Final	589,199	1,772	167,391	495,000

Table 8. Data cleaning steps – Vienna.

	Rental		Sale	
	Observations	Median price	Observations	Median price
Total unique	177,280	899	163,640	320,000
Property type not flat, maisonette, penthouse	9,022	850	10,934	559,000
Square meter $>Q_{97}$	4,958	3,345	4,494	1,620,000
Square meter $<Q_3$	4,368	497	4,090	145,100
price per square meter $>Q_{97}$	4,821	1,498	4,388	999,000
price per square meter $<Q_3$	4,815	550	4,384	179,000
Final	151,176	898	137,650	299,990

6.3. Appendix – Alternative Pre-Crisis Definitions

6.3.1. List-Price Change

Table 9. Price change since december 2018.

	List-price		London		Vienna	
			Sale	Rent	Sale	Rent
Development since Dec. 2018	To pre-crisis definition	1.	3.10%	4.19%	6.60%	−0.54%
		2.	3.46%	4.50%	6.39%	−0.63%
		3.	2.72%	3.97%	5.78%	0.03%
	Until end of 2020		2.74%	−3.50%	13.47%	3.80%
Development during COVID-19 crisis			→	↘	↗	↗

Note: We include three possible "pre-COVID-19 crisis" definitions here: In definition 1 the pre-crisis period covers January and February 2020, in definition 2 the period is expanded to include January, February, and March 2020, and in definition 3 we include the period between October 2019 and March 2020. Mean values over these months are displayed. Similarly, we define the three months of our data sets (October to December 2020) as the "end of 2020".

6.3.2. List-Price-Rent Ratio

Table 10. Change of list-price-rent ratio since january 2019.

	List-price to rent ratio		London	Vienna
Development since Jan. 2019	To pre-crisis definition	1.	−0.58%	6.16%
		2.	−0.17%	6.06%
		3.	−0.23%	4.49%
	Until end of 2020		7.41%	6.58%
Development during COVID-19 crisis			↗	↗

Note: We include three possible "pre-COVID-19 crisis" definitions here: In definition 1 the pre-crisis period covers January and February 2020, in definition 2 the period is expanded to include January, February, and March 2020, and in definition 3 we include the period between October 2019 and March 2020. Mean values over these months are displayed. Similarly, we define the three months of our data sets (October to December 2020) as the "end of 2020".

6.3.3. List Volume Development

Table 11. Percentage list volume change compared to december 2018.

	List volume		London		Vienna	
			Sale	Rent	Sale	Rent
Development since Dec. 2018	To pre-crisis definition	1.	42.78%	37.59%	− 39.71%	− 12.46%
		2.	39.82%	39.33%	− 40.09%	− 11.42%
		3.	24.07%	23.25%	− 26.80%	3.91%
	Until end of 2020		103.00%	96.49%	− 34.39%	49.74%
Development during COVID-19 crisis			↗	↗	→	↗

Note: We include three possible "pre-COVID-19 crisis" definitions here: In definition 1 the pre-crisis period covers January and February 2020, in definition 2 the period is expanded to include January, February, and March 2020, and in definition 3 we include the period between October 2019 and March 2020. Mean values over these months are displayed. Similarly, we define the three months of our data sets (October to December 2020) as the "end of 2020".

6.3.4. List Time-on-Market

Table 12. Average time-on-market in days.

	Monthly time-on-market		London		Vienna	
			Sale	Rent	Sale	Rent
Development since Dec. 2018	To pre-crisis definition	1.	17.99	6.63	20.76	12.79
		2.	18.48	6.24	20.02	12.82
		3.	19.84	6.86	18.33	12.97
	Until end of 2020		12.30	3.53	23.53	12.57
Development during COVID-19 crisis			↘	↘	↗	→

Note: We include three possible "pre-COVID-19 crisis" definitions here: In definition 1 the pre-crisis period covers January and February 2020, in definition 2 the period is expanded to include January, February, and March 2020, and in definition 3 we include the period between October 2019 and March 2020. Mean values over these months are displayed. Similarly, we define the three months of our data sets (October to December 2020) as the "end of 2020".

6.3.5. List Turnover Rate

Table 13. Percentage average turnover.

	List turnover		London		Vienna	
			Sale	Rent	Sale	Rent
Development since Dec. 2018	To pre-crisis definition	1.	62.33%	83.92%	63.98%	78.28%
		2.	61.19%	84.39%	62.09%	76.36%
		3.	57.09%	81.35%	62.53%	75.42%
	Until end of 2020		71.81%	92.07%	75.89%	75.42%
Development during COVID-19 crisis			↗	↗	↗	↗

Note: We include three possible "pre-COVID-19 crisis" definitions here: In definition 1 the pre-crisis period covers January and February 2020, in definition 2 the period is expanded to include January, February, and March 2020, and in definition 3 we include the period between October 2019 and March 2020. Mean values over these months are displayed. Similarly, we define the three months of our data sets (October to December 2020) as the "end of 2020".

Table 14 presents the monthly development of the three liquidity measurements – *volume*, *time-on-market*, and *list turnover rate* – in one table. Each monthly indicator value is compared to the average value of the same indicator over the previous four months. Changing this moving average to include three or five months does not change the result. We assign an upward liquidity trend for a month if two out of the three liquidity indicators indicate a positive market trend (i.e., increasing *volume*, decreasing *time-on-market*, or increasing *list turnover rate*).

Table 14. Monthly liquidity indicators for London and Vienna from march to december 2020 – sales market.

Month	London				Vienna			
(2020)	Volume	TOM	Turnover	Liquidity	Volume	TOM	Turnover	Liquidity
Mar.	↗	↘	↗	(+)	↘	↘	↘	(−)
Apr.	↘	↗	↘	(−)	↘	↗	↘	(−)
May	↘	↗	↗	(−)	↗	↘	↗	(+)
Jun.	↗	↘	↗	(+)	↗	↘	↘	(+)
Jul.	↗	↘	↗	(+)	↗	↘	↘	(+)
Aug.	↗	↘	↗	(+)	↗	↗	↘	(−)
Sep.	↗	↘	↘	(+)	↗	↗	↗	(+)
Oct.	↗	↘	↗	(+)	↘	↗	↗	(−)
Nov.	↘	↘	↘	(−)	↘	↗	↗	(−)
Dec.	↘	↗	−	(−)	↘	↘	−	(+)

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